HabitWork

Minimalist Habit Tracker

"Build habits that actually stick."

Project Report | Portfolio Submission

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Live Website	https://praveengurjar23.github.io/Minimalist-Habit-Tracker/habit-tracker.html

1. Project Overview

HabitWork is a client-side, privacy-first habit tracking web application built entirely with HTML, CSS, and Vanilla JavaScript. It requires no server, no installation, and no account — users simply open a single HTML file in any browser and immediately start tracking their daily habits with full analytics support.

The core philosophy behind HabitWork is to remove every possible barrier between a user and their habits. By eliminating cloud dependency, login flows, and complex setup, the application focuses entirely on what matters: helping users build and maintain daily routines through visual feedback and gamified streak logic.

2. Problem Statement

Most existing habit trackers require account creation, internet connectivity, and subscription fees. First-time users are often discouraged by onboarding complexity before they even track a single habit. HabitWork addresses these pain points directly:

- No signup or login required — open and use instantly
- No internet dependency after the initial page load
- No data leaves the user's device — complete privacy
- Zero cost — fully open source and self-hostable
- Single file architecture — easy to share, fork, and extend

3. Technical Architecture

The entire application is contained within a single HTML file (~1,045 lines). This intentional single-file design ensures portability and zero dependency issues.

Category	Technology	Details	Purpose
Frontend	HTML5 / CSS3	CSS Variables, Flexbox, Grid	Responsive UI & Theming
Logic	Vanilla JS (ES6+)	Custom streak algorithms	State management & logic
Visualization	Chart.js v4.4.1	Bar & line charts	Analytics & trends
Storage	Local Storage API	Client-side persistence	Privacy-first data
Fonts	Google Fonts	Outfit + DM Sans	Typography
Deployment	GitHub Pages	Static site via Actions CI/CD	Live hosting

4. Key Features

Feature	Description	Implementation
Streaks	Tracks current & all-time best streaks per habit	Custom JS algorithm on date keys
Analytics	14-day completion trend & weekday success rate	Chart.js bar + line charts
Calendar	Weekly & monthly views for pattern recognition	Dynamic grid rendering with JS
Confetti	Animated celebration on completing all habits	Canvas-based confetti engine
Toasts	Real-time notifications for habit actions	Timed DOM injection
Privacy	All data stored locally, zero cloud dependency	Local Storage API

5. System Design

5.1 Data Model

All habit data is stored in the browser's Local Storage as a JSON array. Each habit object follows this structure:

```
{ id: "timestamp", name: "Exercise", emoji: "🏋" }
```

Completion records are stored as separate key-value pairs keyed by habit ID and date string (YYYY-MM-DD), allowing O(1) lookups for any given day.

5.2 Streak Algorithm

The streak calculation works by iterating backwards from today's date, counting consecutive days where the habit was marked complete. The algorithm handles time zone-safe date comparisons and correctly distinguishes between a broken streak and a habit not yet marked for today.

5.3 Deployment Pipeline

The repository uses a GitHub Actions workflow (static.yml) to automatically deploy the latest version of habit-tracker.html to GitHub Pages on every push to the main branch. This provides a zero-configuration CI/CD pipeline.

6. Project Structure

File / Directory	Description
habit-tracker.html	Main application file — contains all HTML, CSS, and JS
README.md	Project documentation with tech stack, features, and setup guide
.github/workflows/static.yml	GitHub Actions CI/CD pipeline for auto-deployment to GitHub Pages

7. Future Roadmap

- Data Portability — JSON Export/Import for cross-device migration
- Dark Mode — System-level theme detection and toggle
- Habit Categories — Group habits into Health, Career, and Personal sectors
- Reminder Notifications — Browser push notifications for daily habit reminders
- PWA Support — Progressive Web App for offline mobile use

8. Learning Outcomes

This project was built as a first-year B.Tech portfolio project and demonstrates the following skills:

- Proficiency in Vanilla JavaScript, including ES6+ features, closures, and date manipulation
- CSS architecture using custom properties (variables) for scalable theming
- Integration of third-party libraries (Chart.js) via CDN
- Browser storage API (Local Storage) for client-side data persistence
- CI/CD pipeline setup using GitHub Actions for automated deployment
- Responsive web design principles for cross-device compatibility
- Git version control and open-source project documentation

9. Conclusion

HabitWork demonstrates that a fully functional, analytics-rich web application can be built without any backend infrastructure, build tools, or external dependencies beyond a CDN-hosted charting library. The project prioritises user privacy, zero friction onboarding, and genuine utility — values that are core to good software design.

The application is live, open source, and available for anyone to use, fork, or contribute to. It reflects a practical understanding of frontend web development fundamentals and serves as a foundation for more complex projects in future semesters.